2. *Engine (piston engine)*: The main power source, typically a single-cylinder internal combustion engine.
3. *Fuselage*: The central body of the drone housing the engine, fuel system, and control electronics.
4. *Fixed wings*: Generated lift through aerodynamic design, enabling sustained flight.
5. *Tail assembly (vertical and horizontal stabilizers)*: Provided stability and control in pitch and yaw.
6. *Radio receiver and antenna*: Received control signals from the ground operator, converting them into mechanical actions.
7. *Mechanical control linkages (servomechanisms)*: Translated radio commands into physical movements of control surfaces.
8. *Landing gear*: Allowed take-off and landing on simple runways or open terrain.
9. *Fuel tank*: Stored a small quantity of fuel to power short-duration missions.

2.4.1.3 Operational concept

The operator transmitted signals using a handheld radio controller, guiding the drone in real-time. The onboard receiver converted these signals into servo movements, which controlled flight direction. The drone's propulsion system maintained constant thrust, while the operator adjusted flight attitude through rudder and elevator inputs. Stability was achieved through constant manual corrections, as no autopilot or gyroscopic systems existed at the time.

Figure 2.2 illustrates the fundamental design and operational principles of a first-generation fixed-wing drone, highlighting its structural elements and control mechanisms. These early unmanned aerial vehicles were manually operated and relied on basic radio communication systems to